

JEE MAIN REVISION MASTER: MAGNETIC PROPERTIES OF MATTER SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 73 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Dipole Moment & Earth's Magnetism	39 PYQs	Dipoles & Terrestrial
Very High	Magnetic Materials (Dia, Para, Ferro) & Hysteresis	35 PYQs	Materials

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Dipole Moment & Earth's Magnetism	$\vec{M} = \vec{A} = N I A$, $\mu_B = \frac{e\hbar}{2m} \approx 9.27 \times 10^{-24} \text{ A m}^2$, $B_H = B \cos \delta$, $\tan \delta = \frac{B_V}{B_H}$
Magnetic Materials (Dia, Para, Ferro) & Hysteresis	$\vec{B} = \mu_0 (\vec{H} + \vec{M})$, $\mu_r = 1 + \chi$, $\chi_{\text{para}} = \frac{C}{T}$, $\chi_{\text{ferro}} = \frac{C}{T - T_c}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Dipole Moment & Earth's Magnetism (Very High Yield — 39 PYQs)

Core Principle: Current loops as magnetic dipoles, orbital electron magnetic moment, and elements of Earth's magnetic field.

■■ **Caution / JEE Pitfall:** At the magnetic equator, dip angle $\delta = 0^\circ$ ($B_V = 0$). At magnetic poles, dip angle $\delta = 90^\circ$ ($B_H = 0$).

■ **Speed Trick:** Apparent dip δ_1, δ_2 in two mutually perpendicular vertical planes: $\cot^2 \delta = \cot^2 \delta_1 + \cot^2 \delta_2$.

Representative Source PYQs:

- [2025 Evening Shift] Q#2 (MCQ): "A magnetic dipole experiences a torque of $80\sqrt{3} \text{ Nm}$ when placed in uniform magnetic field in such a way that dipole moment makes angle ..." [Ans: N/A]
- [2025 Evening Shift] Q#4 (MCQ): "Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A): Magnetic monopoles..." [Ans: N/A]
- [2025 Evening Shift] Q#4 (MCQ): "Match List - I with List - II. List - I List - II (A) Magnetic induction (I) Ampere meter 2 (B) Magnetic intensity (II) Weber (C) Magnetic f..." [Ans: A]

3.2 Magnetic Materials (Dia, Para, Ferro) & Hysteresis (Very High Yield — 35 PYQs)

Core Principle: Classification of magnetic materials, Curie-Weiss law, and hysteresis loop area (energy dissipation).

■ **Caution / JEE Pitfall:** Diamagnetic materials have small negative susceptibility $\chi < 0$ that is strictly independent of temperature.

■ **Speed Trick:** Permanent magnets require high retentivity and high coercivity (e.g. Alnico). Transformer cores require high permeability and low hysteresis loss (soft iron).

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . T..." **[Ans: B]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). Consider a ferromagnetic material ..." **[Ans: D]**
- **[2025 Morning Shift] Q#1 (MCQ):** "The percentage increase in magnetic field (B) when space within a current carrying solenoid is filled with magnesium (magnetic susceptibilit..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#2)

A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity in SI units is $\alpha \times 10^5$. The value of α is ____ . (Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units.)

- (A) $(25 / \pi)$
- (B) $(1 / 16\pi)$ **[Correct]**
- (C) $(1 / \pi)$
- (D) $(1 / 4\pi)$

Correct Answer: B

Solution Analysis: The relative permeability of the core is $\mu_r = 400$. The magnetic field applied is, $B = 1.0\text{ T}$. The permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$. The magnetic induction B in a material is related to the magnetic intensity H by the total permeability of the medium (μ) : $B = \mu H$... (i) The total permeability μ is the product of the permeability of free space and the relative permeability of the material : $\mu = \mu_0 \mu_r$ Substituting this into the first equation : $B = \mu_0 \mu_r H \Rightarrow H = \frac{B}{\mu_0 \mu_r}$ Substituting

Benchmark Problem 2: JEE Main 2025 (Online) 29th January Evening Shift (Q#4)

Match List - I with List - II. List - I List - II (A) Magnetic induction (I) Ampere meter (B) Magnetic intensity (II) Weber (C) Magnetic flux (III) Gauss (D) Magnetic moment (IV) Ampere meter Choose the correct answer from the options given below:

- (A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) **[Correct]**
- (B) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
- (C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
- (D) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Correct Answer: A

Solution Analysis: (A) The unit "gauss" is used to measure magnetic induction, also known as magnetic flux density, within the CGS system of units. $B = (\phi / A) = \frac{\text{maxwell}}{\text{cm}^2} = \text{gauss (CGS)}$ (B) Magnetic intensity (H) is defined as the product of the number of turns per unit length in a coil (n) and the current that it carries (I). i.e. $H = nI = (N / L)I$ Unit : A/m or Am^{-1} (C) Magnetic flux, $\phi = BA$ unit : weber (D) Magnetic moment (M) : magnetic moment of a current loop, $M = NIA$ where, N = number of turns I = current A = Area So, unit = Am^2 Hence, option A is correct.